



Energy and Biochar from Sericulture Residues: Briquetting, Pyrolysis and Benefits



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Abstract

Sericulture generates substantial quantities of organic residues such as mulberry prunings, leaf litter, silkworm bed refuse, pupal waste, reeling waste, and other by-products. These materials are often underutilized or disposed of in ways that create environmental and sanitation problems. With increasing emphasis on renewable energy, waste valorisation, and circular bioeconomy models, sericulture residues offer significant potential as feedstock for solid biofuels and biochar. Technologies such as briquetting and pyrolysis enable the conversion of low-density, heterogeneous residues into high-energy fuel and stable carbon-rich soil amendments. This review synthesizes available research on the characterization of sericulture residues, principles of briquetting and pyrolysis, energy recovery pathways, and agronomic and environmental benefits of biochar derived from such biomass. The article highlights how residue-based energy systems can support rural livelihoods, reduce dependence on fossil fuels, improve soil health, and enhance sustainability of the sericulture value chain. Key challenges related to feedstock variability, processing costs, emissions, and policy integration are also discussed, along with future research and development needs.

Keywords: Sericulture residues, briquetting, pyrolysis, biochar, renewable energy, circular bioeconomy

Introduction

Sericulture is an important agro-based rural industry that integrates mulberry cultivation, silkworm rearing, cocoon production, reeling, and silk processing. While the industry is primarily valued for silk, it also generates large volumes of organic residues at every stage of production. These include mulberry pruning waste, leaf litter, unused or leftover leaves, silkworm bed refuse (mixture of excreta, leaf fragments, and moulting skins), dead larvae and pupae, reeling waste, and silk-processing by-products. Traditionally, many of these materials are either dumped, burnt in open fields, or left to decompose in an unmanaged manner, which can cause odor problems, attract pests, and contribute to greenhouse gas emissions (Nagalakshmi *et al.*, 2011). As sustainability and waste valorisation become central to agricultural and industrial policy, attention is increasingly being directed toward converting such residues into useful products rather than treating them as waste.

Biomass-based energy and soil amendments provide two major pathways for value addition. Technologies such as briquetting convert loose, low-density residues into compact solid fuels suitable for domestic cooking, industrial boilers, and decentralized energy systems (Grover & Mishra, 1996). Pyrolysis, a thermochemical process conducted in limited or no oxygen, transforms biomass into bio-oil, syngas, and biochar, a carbon-rich material that can improve soil properties and sequester carbon (Bridgwater, 2012; Lehmann *et al.*, 2011). When applied to sericulture residues, these technologies create opportunities for renewable energy generation, climate change mitigation and enhanced soil fertility, while simultaneously addressing waste management challenges. This review critically examines the characteristics of sericulture residues, the principles and performance of briquetting and pyrolysis technologies, and the agronomic, environmental, and socio-economic benefits of biochar and energy derived from this biomass.

Nature and Availability of Sericulture Residues

Sericulture residues originate from both the agricultural and industrial components of silk production. On the farm side, mulberry cultivation generates significant quantities of pruned branches, twigs, senescent leaves, and field litter during routine maintenance and harvesting. In rearing houses, silkworm bed refuse accumulates after each instar and consists of partially eaten leaves, faecal matter, exuviae, and microbial biomass. After cocoon harvesting and reeling, additional residues such as pierced cocoons, silk floss, and pupal waste are produced. Depending on the scale of sericulture operations, these residues can amount to several tonnes per hectare annually.

From a biomass utilization perspective, these materials possess moderate calorific value and high organic carbon content, but they are often bulky, heterogeneous, and seasonally variable. Their low bulk density, high moisture content, and fibrous nature limit direct use as fuel or soil amendment without pre-processing (Kaliyan & Morey, 2009). Nevertheless, their lignocellulosic composition makes them suitable feedstock for densification and thermochemical conversion processes. Integrating residue management into sericulture systems not only reduces environmental burden but also enhances the economic resilience of farmers and reeling units by generating additional value streams (Nagalakshmi *et al.*, 2011).

Briquetting of Sericulture Residues for Energy

Briquetting is a mechanical densification process in which loose biomass is compressed under high pressure into uniform, solid blocks or cylinders with improved handling, storage, and combustion characteristics. Agricultural residues such as straw, husk, stalks, and pruning wastes have been widely studied for briquette production, and similar principles apply to sericulture residues (Grover & Mishra, 1996). The primary objectives of briquetting are to increase bulk density, reduce transportation and storage costs, and enhance combustion efficiency.

Sericulture residues such as mulberry twigs, dried leaves, and rearing litter can be shredded, dried, and compacted into briquettes either alone or in combination with other agro-wastes to improve binding and fuel quality. The quality of briquettes is influenced by moisture content, particle size, compaction pressure, and the presence of natural binders such as lignin (Kaliyan & Morey, 2009). Properly manufactured briquettes exhibit higher calorific value per unit volume, uniform burning characteristics, and lower smoke emission compared to loose biomass. Studies on agro-waste briquetting demonstrate that densified fuels can effectively substitute for firewood and coal in rural households and small industries, thereby reducing deforestation and fossil fuel consumption (Sahay & Singh, 2014).

When applied in sericulture regions, briquetting offers multiple benefits. First, it provides an on-farm or community-level energy source for cooking, space heating, and process heat in reeling units. Second, it reduces the need for open burning or dumping of residues, minimizing air pollution and sanitation issues. Third, briquet production can become a small-scale enterprise for farmer groups, cooperatives, and self-help groups, generating additional income and employment. However, challenges include the need for drying infrastructure, initial investment in briquetting equipment, and the development of reliable residue collection systems.

Pyrolysis of Sericulture Biomass

Pyrolysis is a thermochemical conversion process in which biomass is heated in the absence or near-absence of oxygen, resulting in the production of three main products: bio-oil (liquid), syngas (gaseous), and biochar (solid). Depending on temperature, heating rate, and residence time, pyrolysis can be classified as slow, fast, or intermediate, each favouring different product distributions (Mohan *et al.*, 2006; Bridgwater, 2012). For agricultural and forestry residues, slow pyrolysis is commonly used to maximize biochar yield, while fast pyrolysis emphasizes bio-oil production.

Sericulture residues, being lignocellulosic in nature, are well suited for pyrolysis. Mulberry wood, twigs, leaf litter, and rearing waste can be converted into energy-rich gases and stable biochar. The gaseous and liquid fractions can be utilized as renewable energy sources for heating or power generation, while the biochar

fraction represents a long-term carbon sink and soil amendment. Pyrolysis also offers advantages in terms of waste volume reduction and pathogen destruction, making it attractive for managing organic residues from rearing houses and processing units.

Thermochemical studies on agricultural residues indicate that pyrolysis efficiency and product quality depend strongly on feedstock composition and operating parameters (Mohan et al., 2006). For example, higher lignin content generally favours char formation, while higher cellulose and hemicellulose content increase volatile production. Applying this understanding to sericulture biomass allows optimization of process conditions to balance energy recovery and biochar yield. Although large-scale pyrolysis systems require significant capital investment, small and modular units suitable for rural contexts are increasingly being developed, making decentralized biomass conversion more feasible.

Biochar from Sericulture Residues: Properties and Soil Benefits

Biochar is the carbon-rich solid residue produced during pyrolysis and has gained global attention for its potential to improve soil health, enhance nutrient use efficiency, and sequester carbon (Lehmann *et al.*, 2011). Compared to raw organic matter, biochar is more chemically stable and can persist in soils for decades to centuries, thereby contributing to long-term carbon storage (Spokas *et al.*, 2012).

Biochar derived from sericulture residues exhibits porous structure, high surface area, and variable pH depending on production conditions. When applied to soil, such biochar can improve physical properties by increasing water-holding capacity and aggregation, particularly in light-textured or degraded soils (Liu *et al.*, 2018). Chemically, biochar enhances cation exchange capacity, reduces nutrient leaching, and can immobilize contaminants, thereby improving nutrient availability and environmental quality (Ahmad *et al.*, 2014). Biologically, biochar provides habitat for soil microorganisms and can stimulate beneficial microbial activity, indirectly supporting plant growth.

Field studies on biochar from various agricultural residues show positive effects on crop productivity and soil fertility over multiple seasons. For instance, long-term experiments have demonstrated improved maize yield, nutrient uptake, and soil organic carbon following biochar application (Major *et al.*, 2010). Although specific studies on mulberry or sericulture-waste biochar remain limited, the general principles observed in other biomass-derived biochars are applicable. In sericulture systems, applying biochar to mulberry fields could improve soil moisture retention, nutrient efficiency, and root development, ultimately enhancing leaf yield and quality for silkworm rearing. Furthermore, integrating biochar with composted sericulture waste may create synergistic effects, combining immediate nutrient supply with long-term soil conditioning.

Environmental and Climate Change Implications

The conversion of sericulture residues into energy and biochar has significant environmental benefits. Briquetting and pyrolysis reduce the need for open burning and uncontrolled decomposition, both of which contribute to air pollution and greenhouse gas emissions. Substituting biomass-based fuels for coal, firewood, or fossil fuels lowers carbon dioxide emissions associated with energy use. More importantly, biochar application represents a form of negative emission technology, as it transfers atmospheric carbon fixed by plants into stable soil carbon pools (Lehmann *et al.*, 2011).

Biochar also plays a role in mitigating soil degradation and nutrient losses. By improving nutrient retention and reducing leaching of nitrogen and phosphorus, biochar can decrease fertilizer requirements and associated environmental impacts such as eutrophication (Spokas et al., 2012). In sericulture regions, where mulberry cultivation often occurs on marginal or intensively managed lands, such soil improvements are particularly valuable. Additionally, biochar's capacity to adsorb agrochemicals and heavy metals offers opportunities for remediation of contaminated soils and effluents from silk processing units (Ahmad *et al.*, 2014).

Socio-Economic Potential and Rural Livelihoods

From a socio-economic perspective, residue-based energy and biochar systems can strengthen rural livelihoods and promote decentralized development. Briquette production and small-scale pyrolysis units can be established at village or cluster levels, creating employment in residue collection, processing, equipment operation, and product marketing. For sericulture farmers, these activities provide additional income streams beyond cocoon sales and reduce expenditure on conventional fuels and soil amendments.

Integrating energy and biochar production into sericulture cooperatives and self-help groups can enhance collective action and local entrepreneurship. Moreover, the use of biochar in mulberry cultivation can increase leaf productivity and resilience under moisture stress, indirectly improving cocoon yield and farm profitability. Such integrated models align closely with the principles of circular economy, where waste from one component becomes a resource for another, enhancing overall system efficiency (Van Huis *et al.*, 2013).

Constraints and Research Gaps

Despite promising potential, several challenges must be addressed for large-scale adoption. The heterogeneous nature of sericulture residues leads to variability in fuel quality and biochar properties, necessitating standardization of feedstock preparation and processing parameters. Initial capital investment for briquetting and pyrolysis equipment can be a barrier for smallholders, highlighting the need for institutional support, subsidies, and cooperative models. Technical issues such as moisture control, emissions management, and quality assurance of biochar also require attention.

Research gaps include limited data on the long-term agronomic effects of biochar specifically derived from sericulture residues, optimal application rates for mulberry, and the integration of biochar with organic and inorganic fertilization regimes. Life-cycle assessments comparing residue-based energy systems with conventional waste management practices would further clarify environmental benefits. Finally, policy frameworks that recognize biochar and biomass energy as components of climate-smart agriculture and renewable energy strategies are essential for scaling up these technologies.

Conclusion

Sericulture residues represent an underutilized biomass resource with significant potential for renewable energy generation and soil improvement through briquetting and pyrolysis. Briquetting transforms bulky wastes into efficient solid fuels, supporting rural energy needs and reducing dependence on fossil fuels. Pyrolysis enables the simultaneous production of bioenergy and biochar, offering a pathway for climate change mitigation, waste reduction, and soil rehabilitation. Biochar derived from sericulture biomass can enhance soil physical, chemical, and biological properties, improve mulberry productivity, and contribute to long-term carbon sequestration. Although technical, economic, and institutional challenges remain, integrating energy and biochar production into sericulture systems offers a sustainable, circular approach that strengthens environmental performance and rural livelihoods. Continued research, standardization, and supportive policy interventions will be crucial to fully realize the potential of sericulture residues in renewable energy and sustainable agriculture.

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